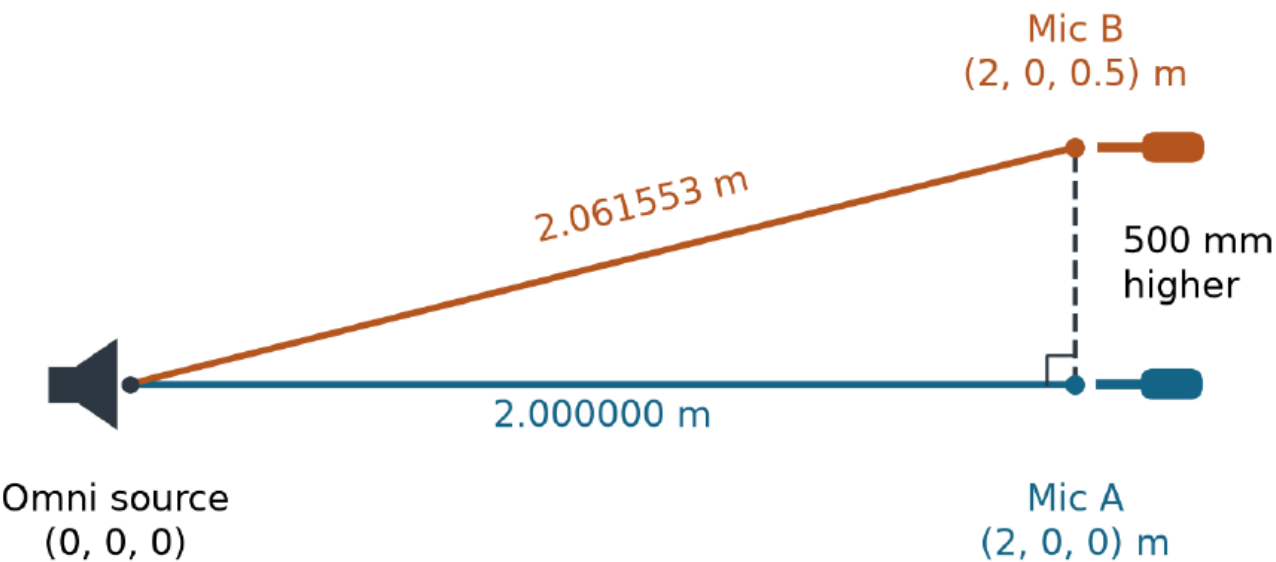


Does HALS preserve sound travel time

A simple Stage 5 test with a synthetic point source

Move a virtual microphone upwards by 500 mm. Sound now travels along a slightly longer diagonal path. Does the exported phase show exactly the extra delay we expect?



Coordinates R, Phi, Z: Here both microphones have R = 2 m and their heights are offset by 0.5 m.

Table 1 Effect of the 500 mm height offset

Values describe the change from Mic A to Mic B (B minus A).

Quantity	Geometry predicts	HALS export measures
Path length	61.552813 mm	61.552839 mm
Delay	179.454265 μ s	179.454341 μ s
Phase slope per kHz	-64.603535°	-64.603563°

The delay error is: 0.000076 μ s, equivalent to **0.000026 mm** of sound travel.

Largest phase error: 0.031° across 20 Hz to 20 kHz.

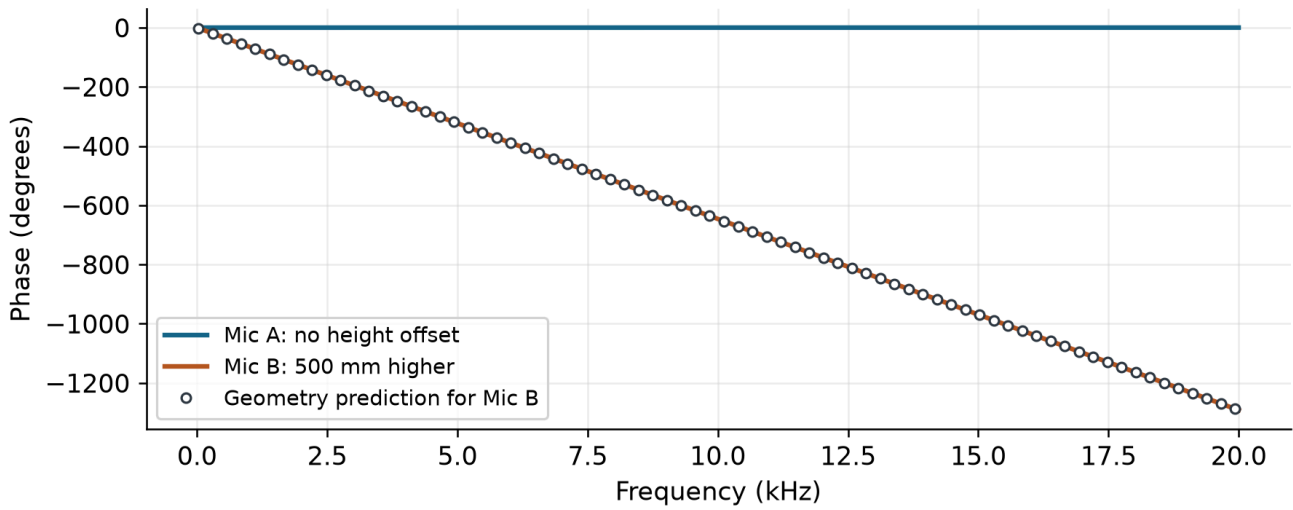
The geometry check in three steps

- 1. Distance:** Pythagoras gives $\sqrt{2^2 + 0.5^2} = 2.061552813$ m.
- 2. Travel time:** $2.061552813 \div 343 = 6010.358055$ μ s.
- 3. Extra travel time relative to Mic A:**
 $(2.061552813 - 2) \div 343 = 179.454265$ μ s.

At 1 kHz, one cycle takes 1 ms and represents 360°. The extra delay is 0.179454 of that cycle, giving **64.6035° of phase lag per kHz** after subtracting the common 2 m travel time.

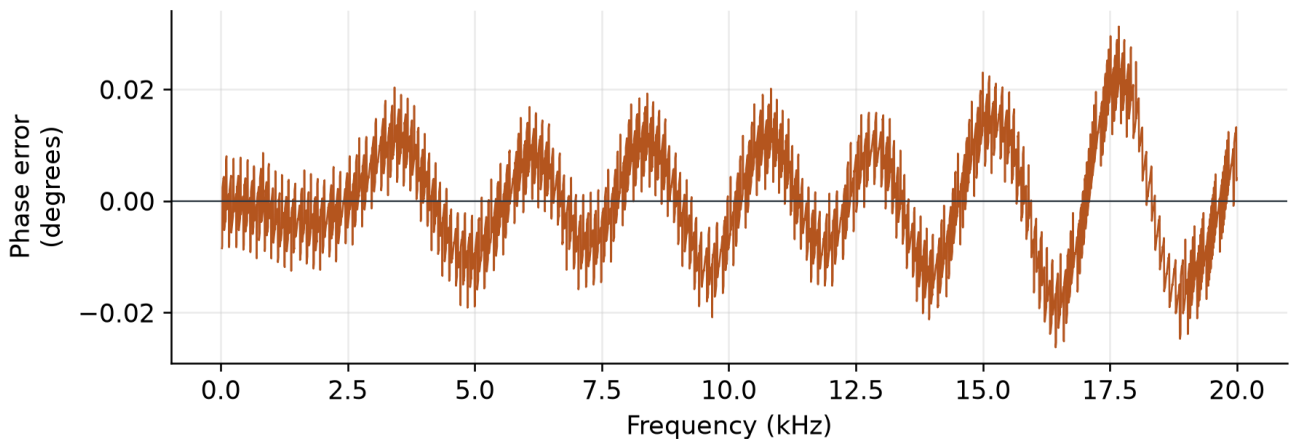
The phase plots

Ref Origin TOF subtraction mode removes the same 2 m travel time from both microphones. Mic A then sits near zero phase. Mic B keeps the extra delay caused by its greater distance. Its phase follows the geometric prediction.



Mic B and the prediction overlap. The downward slope is the expected extra travel time.

A closer look at the difference



The exported phase difference between Mic B, minus the geometric prediction. The vertical scale is magnified to reveal the remaining error.

What does the tiny difference mean

Across 20 Hz to 20 kHz, the error in the extra travel time is **0.000076 μ s** (about 0.076 nanoseconds), equivalent to **0.000026 mm** of travel. The generator, solved model and exporter all use **343 m/s**, so this is not a mismatch between their sound-speed settings.

The small remaining difference can include sampling, model-fitting and export-rounding effects. The match remains close across the full audible range.

The direct complex frequency-domain output (NPZ) and the text response (FRD) give delays that differ by only **0.015 nanoseconds**. Both agree closely with the geometric prediction.

Result: HALS preserves the expected extra sound travel time in this two-position synthetic test.

Test setup and how to repeat it

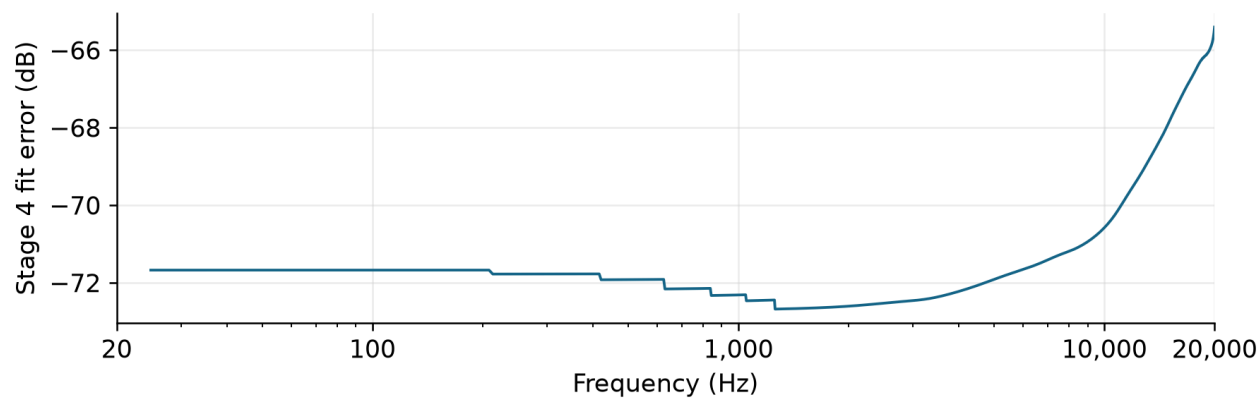
The test folder `src/misc/system_testing/hals_tof_test` already contains `synth_ir_gen_tof_test.py`, configured with radius 0 and one source point. It generates a full range omnidirectional source using the accompanying `synth_ir_grid.csv`, which contains **1,443 measurement positions**.

Setting	Value for this test
Synthetic source	Centre (0, 0, 0) m; radius 0 m; piston count 1
Sound speed	343 m/s in the generator, solve and export
Stage 4 solve	Maximum order N8; origin fixed at (0, 0, 0)
Use optimised origins	Off for the test; no origin shifts in the solved model
Stage 5 Advanced settings	Manual IR capture padding: 0 samples
Stage 5 microphone settings	2 m radius; theta 90°; phi 0°; Z offset 0 then 500 mm
Stage 5 response settings	Internal field; Ref Origin TOF; no mic calibration

Generator: radius zero places the single source point exactly at the origin.

Stage 5: capture padding must be zero because the synthetic generator adds no leading capture padding to the IRs.

How well did Stage 4 fit the synthetic measurements



Saved Stage 4 fit error across the measured grid. Lower is better: -60 dB means the remaining error is 0.1% of the measured response level. This checks the model fit, separately from the travel-time test.

Across 20 Hz to 20 kHz, the median fit error is **0.030%** and the maximum is **0.054%**. N8 is the upper limit; this solve uses orders 2 to 8 as frequency increases.

Repeat the test

Open a terminal in `src/misc/system_testing/hals_tof_test`. Run `synth_ir_gen_tof_test.py` to generate the WAVs, then import them into HALS and solve using the settings above. To test the solved model, run:

```
python export_test.py --coeff "path/to/your_model.h5"
python analyze_test.py
```

Omit `--coeff` to use the existing model in this test folder. The first script exports both microphones; the second creates the metrics and all report graphs. See the local README for the full instructions.